LAB 1

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B-11

1. PROGRAM TO CALCULATE SIMPLE INTEREST.

Program_1.c > main()

```c
#include <stdio.h>
int main()
{
    int p, n, r, si;

    printf("Enter the value of principal = ");
    scanf("%d", &p);

    printf("Enter the value of rate of interest = ");
    scanf("%d", &r);

    printf("Enter the value of year = ");
    scanf("%d", &n);

    si = (p*r*n)/100;

    printf("Enter the value of Simple Interest = %d", si);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) {
Enter the value of principal = 1000
Enter the value of rate of interest = 5
Enter the value of year = 2
Enter the value of Simple Interest = 100
PS C:\Users\Lenovo\Desktop\C>
```

2. PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL.

Program_2.c > main()

```c
#include <stdio.h>
int main()
{
    int phy, maths, chem, eng, mar, total;
    float percentage;

    printf("Enter the value of phy = ");
    scanf("%d", &phy);
    printf("Enter the value of maths = ");
    scanf("%d", &maths);
    printf("Enter the value of chem = ");
    scanf("%d", &chem);
    printf("Enter the value of eng = ");
    scanf("%d", &eng);
    printf("Enter the value of mar = ");
    scanf("%d", &mar);
    total = phy + maths + chem + eng + mar;
    percentage = total/5.0;
    printf("Enter the value of total = %d", total);
    printf("Enter the value of percentage = %f", percentage);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { g
Enter the value of phy = 100
Enter the value of maths = 94
Enter the value of chem = 93
Enter the value of eng = 92
Enter the value of mar = 98
Enter the value of total = 477Enter the value of percentage = 95.400002
PS C:\Users\Lenovo\Desktop\C>
```

3. PROGRAM TO CALCULATE GROSS SALARY.

Program_3.c > main()

```c
#include <stdio.h>
int main()
{
    int salary, allowances, bonuses, total;

    printf("Enter the salary : ");
    scanf("%d", &salary);

    printf("Enter the allowances : ");
    scanf("%d", &allowances);

    printf("Enter the bonuses : ");
    scanf("%d", &bonuses);

    total = salary+allowances+bonuses;

    printf("The Gross Salary = %d", total);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the salary : 50000
Enter the allowances : 20000
Enter the bonuses : 10000
The Gross Salary = 80000
PS C:\Users\Lenovo\Desktop\C>
```

4. PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TO CENTIGRADE.

Program_4.c > main()

```c
#include <stdio.h>
int main()
{
    int f, c;

    printf("Enter the value of f = ");
    scanf("%d", &f);

    c = (f - 32)*5/9;

    printf("The value of c = %d", c);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if (
Enter the value of f = 212
The value of c = 100
PS C:\Users\Lenovo\Desktop\C>
```

5. PROGRAM TO SWAP TWO VARIABLES USING THIRD VARIABLE.

Program_5.c > main()

```c
#include <stdio.h>
int main()
{
    int a, b, temp;

    printf("Enter the value of a = ");
    scanf("%d", &a);

    printf("Enter the value of b = ");
    scanf("%d", &b);

    temp = a;
    a = b;
    b = temp;

    printf("After swapping : a=%d, b=%d", a, b);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($
Enter the value of a = 23
Enter the value of b = 45
After swapping : a=45, b=23
PS C:\Users\Lenovo\Desktop\C>
```

6. PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE.

Program_6.c > main()

```c
#include <stdio.h>
int main()
{
    int a, b;

    printf("Enter the value of a = ");
    scanf("%d", &a);

    printf("Enter the value of b = ");
    scanf("%d", &b);

    a = a + b;
    b = a - b;
    a = a - b;

    printf("After swapping: a=%d, b=%d", a, b);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of a = 13
Enter the value of b = 67
After swapping: a=67, b=13
PS C:\Users\Lenovo\Desktop\C>
```

7. PROGRAM TO CALCULATE AREA OF TRIANGLE.

Program_7.c > main()

```c
#include <stdio.h>

int main()
{
    int base, height;
    float area;

    printf("Enter the base of Triangle:");
    scanf("%d", &base);

    printf("Enter the height of Triangle:");
    scanf("%d", &height);

    area = 0.5 * base * height;

    printf("Area of Triangle = %.2f\n", area);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if (
Enter the base of Triangle:2
Enter the height of Triangle:5
Area of Triangle = 5.00
PS C:\Users\Lenovo\Desktop\C>
```

8. EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS. WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAYS, HOURS AND MINUTES.

```c
#include <stdio.h>
int main()
{
    int sec, min, hr, days;

    sec = 31558150;

   min = sec/60;
   hr = sec/(60*60);
   days = sec/(60*60*24);

   printf("value of min=%d, hr=%d, days=%d", min, hr, days);

   return 0;
}
```

```
Problems  Output  Debug Console  Terminal  Ports

PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?) { gcc P
value of min=525969, hr=8766, days=365
PS C:\Users\Lenovo\Desktop\C>
```

9. PROGRAM TO READ TIME IN HR, MIN, SEC AND CONVERT IT INTO TOTAL SECONDS.

Program_9.c > main()

```c
#include <stdio.h>
int main()
{
    int sec, min, hr, SEC, a, b, c;

    printf("Enter the value of hr, min, sec = ");
    scanf("%d%d%d", &hr, &min, &sec);

    a = hr * 3600;
    b = min * 60;
    c = sec * 1;

    SEC = a + b + c;

    printf("Total SEC = %d", SEC);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if
Enter the value of hr, min, sec = 1 30 30
Total SEC = 5430
PS C:\Users\Lenovo\Desktop\C>
```

10. WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA
    CM = M/2+P/2+C/2+E
    WHERE CM = CUT OFF MARK
    M = Marks in mathematics out of 200
    P = Marks in physics out of 200
    C = Marks in chemistry out of 200
    E = Marks in entrance examination out of 100

Program_10.c > main()

```c
#include <stdio.h>
int main()
{
    int M, P, C, E, CM;

    printf("Enter the value of M=, P=, C=, E=");
    scanf("%d%d%d%d", &M, &P, &C, &E);

    CM = M/2 + P/2 + C/2 + E;

    printf("value of CM = %d", CM);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter the value of M=, P=, C=, E= 200 200 200 100
value of CM = 400
PS C:\Users\Lenovo\Desktop\C>
```

11. PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

Program_11.c > main()

```c
#include <stdio.h>
int main()
{
    int SEC, hr, min, sec, remainingMin, remainingSec;

    printf("Enter the value of SEC = ");
    scanf("%d", &SEC);

    hr = SEC/3600;

    remainingMin = SEC % 3600;

    min = remainingMin/60;

    sec = remainingMin % 60;

    printf("Time is %d:%d:%d", hr, min, sec);

    return 0;
}
```

Problems Output Debug Console Terminal Ports

```
PS C:\Users\Lenovo\Desktop\C> cd "c:\Users\Lenovo\Desktop\C\" ; if ($?)
Enter the value of SEC = 5430
Time is 1:30:30
PS C:\Users\Lenovo\Desktop\C>
```